LAB 2

GEOSPATIAL REPRESENTATION LEARNING

From Geodata to a Multimodal Dataset

Build the data pipeline we will use in Lab 3.

PRESS → OR SPACE. →

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Learning Outcomes

After this lab, you will be able to:

Collect and prepare geospatial data for use in a machine-learning pipeline.

Not comfortable — Very comfortable

Click to respond

Develop a custom PyTorch dataset that retrieves values from one geospatial source at specified coordinates.

Not comfortable — Very comfortable

Click to respond

Contribute to a shared programming project through GitHub, using forks, commits, and pull requests.

Not comfortable — Very comfortable

Click to respond

Our Study Area



Map placeholder

AOI boundary · geographic extent · example query point

[Study area map to be provided]

Study area

[NAME TO BE PROVIDED]

AOI boundary / bounding box

[RESOURCE LINK TO BE PROVIDED]

Shared Drive folder

[LINK TO BE PROVIDED]

Every team collects data covering the same area.

One Coordinate, Many Measurements

Example query

```
dataset.query(latitude=..., longitude=...)
```

```
{
    "land_cover":     ...,
    "building":       ...,
    "your_modality":  ...
}
```

Query interface is illustrative — align with the actual repository when available.

PROVIDED EXAMPLES

🗂️ **Land-cover raster**

containing pixel → class code

🏢 **Building polygons**

point containment → 0 or 1

TEAM CONTRIBUTIONS

Additional assigned modalities — one per team. Teams adapt these examples to their source.

Teams and Data Sources

Team	Data Specialist	Data Loader Specialist	Integration Specialist	Modality / Source
1	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
2	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
3	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
4	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
5	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
6	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
7	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
8	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*
9	*[to be provided]*	*[to be provided]*	*[to be provided]*	*[to be provided]*

Data Specialist

1. Find a suitable source for your team's assigned modality.
2. Download or export a subset covering the shared study area.
3. Inspect the data and confirm what its values represent.
4. Upload the prepared subset to the shared Google Drive folder.
5. Share the file link and essential metadata with your team.
   Variables, units/classes, CRS, observation date, and missing values.

SHARED DRIVE FOLDER

[LINK TO BE PROVIDED]

DELIVERABLE

A checked data subset on Drive, with its source and essential metadata.

Collaborate early: Agree on file format and variables with your Data Loader Specialist.

Data Loader Specialist

1. Fork the supplied repository once for your team and clone it into your working environment.
2. Copy the land-cover or building-footprint dataset as a starting point.
3. Adapt the download script to retrieve your Drive file and cache it in `data/`.
4. Adapt the dataset to return your variables at a requested coordinate.
5. Verify results at known locations and run the provided checks.
6. Commit your changes and push them to your team's fork.

STARTING REPOSITORY

[LINK TO BE PROVIDED]

DELIVERABLE

A working, committed point-query dataset for your modality in your team's fork.

Collaborate: Get data and metadata from your Data Specialist; agree on the interface with your Integration Specialist.

Integration Specialist

1. Inspect the combined dataset script in your team's fork.
2. Understand how it calls individual datasets and combines their outputs.
3. Agree with your team on the interface and variable names.
4. Connect your team's dataset to the combined query and verify it works.
5. Coordinate a pull request to the central repository.
6. Review and integrate contributions with the instructor and other Integration Specialists.

CENTRAL REPOSITORY

[LINK TO BE PROVIDED]

DELIVERABLE

A reviewed pull request connecting your modality to the shared dataset.

Coordinate across teams so all contributions work together.

Collaborating with GitHub

1. **Fork** the instructor's repository — your team's copy on GitHub.
2. **Clone** your team's fork — working copy on your laptop or in Colab.
3. Develop, **commit**, and **push**.
4. Open a **pull request** — proposal to integrate changes into the central repository.
5. Review, revise, and merge.



Your team develops one modality. A pull request brings it into our shared dataset.

The instructor and Integration Specialists review contributions and keep the shared dataset working.

Coding with AI Agents

Available tools (VS Code)

GitHub Copilot
Claude Code
OpenAI Codex

Install and sign into one option before the lab.

WORKING PATTERN

Give context → request a change → inspect → run and verify → commit

EXAMPLE PROMPT

"Read the example datasets. Adapt the appropriate example for our modality, preserve the point-query interface, and run the provided checks."

You are responsible for understanding the data and verifying the result.

The Shared Workflow

DATA SPECIALIST	DATA LOADER SPECIALIST	INTEGRATION SPECIALIST
Find source	Adapt example	Inspect combined dataset
↓	↓	↓
Prepare subset < 50 MB	Connect uploaded data → data/ cache	Connect contribution
↓	↓	↓
Upload to Drive	Verify point queries	Review and merge PR
Drive — prepared data *[LINK TO BE PROVIDED]*	**GitHub** — code & collaboration *[CENTRAL REPO LINK TO BE PROVIDED]*	**Colab notebook** *[LINK TO BE PROVIDED]*

Two Hours, Three Checkpoints

40 min

Data selected and prepared; loader and integration work underway.

80 min

Your dataset returns values from your own data.

120 min

Your contribution is integrated and queried through the shared dataset.

What Counts as Done?

- ✓ Prepared data cover the AOI and are accessible from Drive.
- ✓ Dataset downloads missing files and caches them locally.
- ✓ Coordinate queries return named values with documented meaning.
- ✓ Missing and outside-coverage results are handled explicitly.
- ✓ A pull request contributes the modality to the shared dataset.
- ✓ Checks pass and the combined query works.

Team deliverable: a working source contribution.

Class outcome: a working multimodal dataset.

What We Built Together



One coordinate → multiple modalities → a machine-learning sample

In Lab 3, we will use this dataset to build features and train machine-learning models.